NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-6106/12C(USAF)
15 April 1994
SUPERSEDING
MIL-R-6106/12B(USAF)
22 December 1989

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC (TYPE I), MAGNETIC LATCH, 2 AMPERES, 1PDT, AUXILIARY CONTACTS 25 AMPERES, 3 PNO MAIN CONTACTS, HERMETICALLY SEALED

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this Specification sheet and the issue of the following specification list in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-6106.

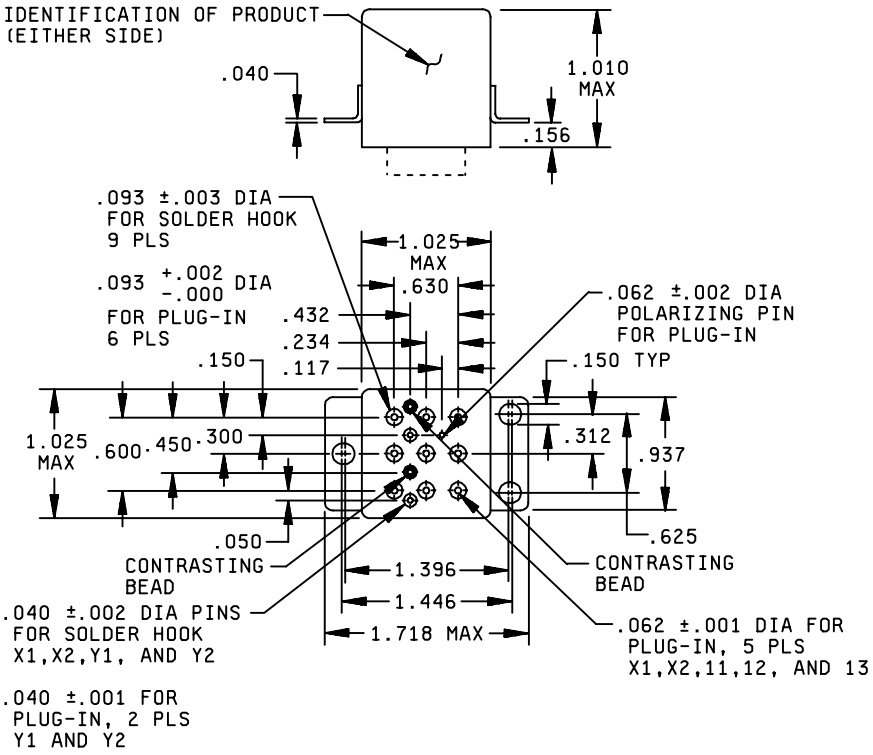
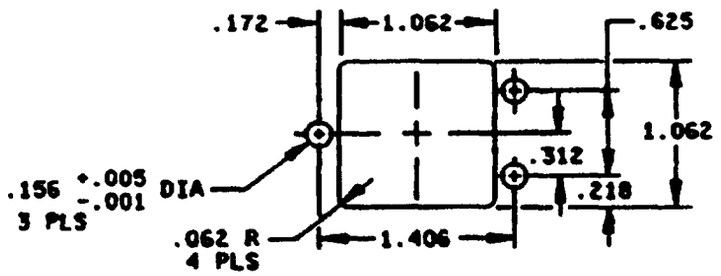


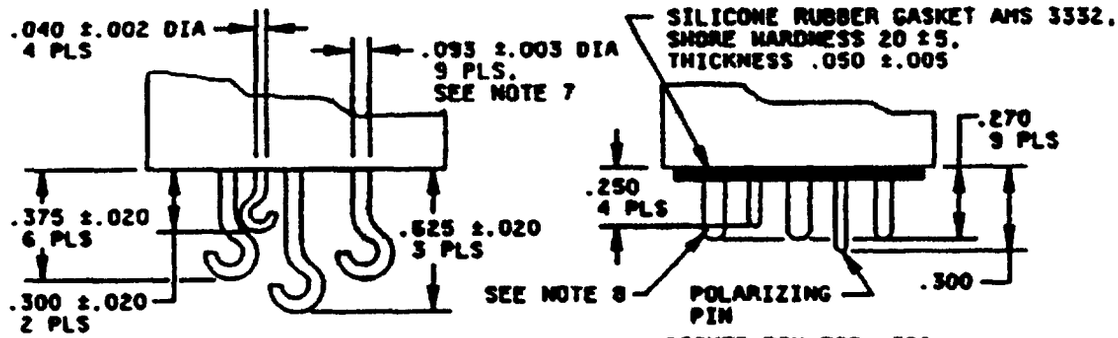
FIGURE 1. Relay, outline drawing.

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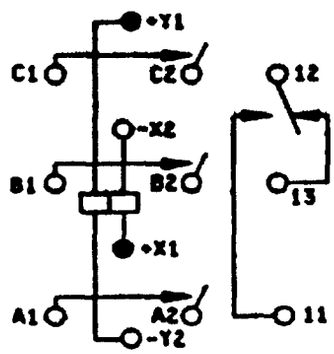


MOUNTING HOLE LAYOUT FOR -001 AND -002



SOLDER HOOK FOR -001
(ALL HOOKS TIN PLATED)

SOCKET PIN FOR -002



CIRCUIT DIAGRAM
TERMINAL VIEW

© FIGURE 1. Relay, outline drawing - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.156	3.96	.600	15.24
.002	0.05	.172	4.37	.625	15.88
.003	0.08	.234	5.94	.630	16.00
.005	0.13	.250	6.35	.937	23.80
.020	0.51	.270	6.86	1.010	25.65
.040	1.02	.300	7.62	1.025	26.04
.050	1.27	.312	7.92	1.062	26.97
.062	1.57	.385	9.78	1.396	35.46
.093	2.36	.395	10.03	1.406	35.71
.117	2.97	.432	10.97	1.446	36.73
.150	3.81	.450	11.43	1.718	43.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified.
5. Notes to observe polarity must appear on relays.
6. To close main contact, (operate) energize X1 and X2. To open main contacts, (reset) energize Y1 and Y2.
7. Hook terminals will accept one no. 12 AWG stranded wire.
8. Pins shall be gold plated in accordance with MIL-G-45204, type II, class I. Underplating: nickel, 50 to 150 microinches thick (gold plating of polarizing pin optional).

© FIGURE 1. Relay, outline drawing - Continued.

REQUIREMENTS:

Coil data: See table I.

Rated contact load: See table II.

Dimensions and configuration: See figure 1.

Weight: 0.18 pound (81.64 grams) maximum.

Temperature range 1/: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Shock G-level: 200 g.

Duration: 6 ms.

Maximum duration contact opening: 10 µs.

1/ For full rated load at maximum temperature and altitude, use no. 12 wire or larger. Solder hook relays are to be mounted to limit mounting bracket temperature to +135°C.

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Vibration-sinusoidal:

0.12 inch DA, 10 Hz - 70 Hz.

30 g, 70 Hz - 3,000 Hz.

Vibration-random:

Applicable specification: MIL-STD-202, method 214, test condition 1B.

Duration: 15 minutes, each plane.

Acceleration: 15 g.

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level): 2 to 5 seconds.

	Initial	After life tests
Coil to case and coil to coil	1,000 V rms	1,000 V rms
Auxiliary contacts	1,000 V rms	750 V rms
All other points	1,250 V rms	1,000 V rms

Dielectric withstanding voltage 2/: 1 minute.

	80,000 feet	300,000 feet
Coil to case and coil to coil	350 V rms	500 V rms
Auxiliary contacts	350 V rms	500 V rms
All other points	350 V rms	500 V rms

Maximum contact drop, initial: 0.150 volt.

After life tests: 0.175 volt.

Overload current: 50 amperes dc, 80 amperes ac.

Rupture current: 60 amperes dc, 100 amperes ac.

Duty rating: Continuous.

Part or Identifying Number (PIN): M6106/12-(plus dash number from table I).

2/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

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© TABLE I. Operating characteristics.

PIN M6106 /12	Type	Coil data								Time (milliseconds, maximum)						Terminals
		Coils	Rated		Maximum		Maximum pickup voltage			Operate	Release	Bounce				
												Main		Aux		
			V dc 1/	Res ±10% at +25°C	V dc	A	Norm 2/	High temp test	Cont current test	3/	3/	NO	NC	NO	NC	
-001	I	X1,X2, Y1,Y2	28	450	29	.075	18	19.8	22.5	15	15	1.0		4	4	Solder hook
-002	I	X1,X2, Y1,Y2	28	450	29	.075	18	19.8	22.5	15	15	1.0		4	4	Socket pin

1/ Caution: Use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.

© TABLE II. Rated contact load (amperes per pole) (case grounded). 1/ 2/ 3/

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz 2/	60 Hz
4/ Resistive	50	25		2	2	25		2		25			
Inductive	10	12											
Inductive	10			1	1	15		1		15			
Motor	50	10				10				10			
Lamp	50	5		.5	.5	5		.5		5			
5/ Transfer load													
Mechanical life reduced current	200	6				6				6			
Intermediate current	Applicable in accordance with MIL-R-6106 (main contacts only)												

1/ Absence of value indicates relay is not rated for 3 phase application.
2/ Single phase load shall be run during the 3 phase test on auxiliary contacts.
3/ Absence of value indicates parameter is not applicable to this specification sheet.
4/ For full rated load at maximum temperature and altitude, use no. 12 wire or larger. Solder hook relays shall be mounted to limit mounting bracket temperature to +135°C.
5/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at the rating indicated.

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CONCLUDING MATERIAL

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F759)

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